# Chloe L. Boehm

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## EDUCATION

**Colorado State University**, Fort Collins, CO, USA 2022-2025
Ph.D., Atmospheric Science
Dissertation: Two-Way Interactions Between Antarctic Sea Ice and the Atmospheric Circulation: Mechanisms for and Implications of Recent Sea-Ice Loss
Advisor: David W.J. Thompson

**Colorado State University**, Fort Collins, CO, USA 2019-2022
M.S., Atmospheric Science
Thesis: The Contribution of Clouds to Global Surface Temperature Variability on Monthly to Decadal Timescales
Advisor: David W.J. Thompson

**Colby College**, Waterville ME, USA 2014-2018
B.A, *cum laude*, Double Major: Physics and Mathematical Sciences, Minor: Computer Science

**University of Otago**, Dunedin, New Zealand Spring 2017

## RESEARCH EXPERIENCE

**Alfred Wegener Institute,** Bremerhaven, Germany February 2026-present
Postdoctoral Researcher, Helmholtz Investigator Group 'Southern Ocean & Antarctic Sea Ice Evolution'
Supervisor: Lettie Roach

**World Climate Research Programme** 2024-present
Explaining and Predicting Earth System Change (EPESC)
Working Group 2: Integrated Attribution, Prediction and Projection

**Colorado State University**, Fort Collins, CO, USA 2019-2025
Graduate Research Assistant
Advisor: David W.J. Thompson

**University of Bergen**, Bergen, Norway September 2022
Visiting Researcher
Supervisor: Shengping He

**Mount Washington Observatory**, North Conway, NH, USA 2018-2019
Summit Internship, Researching novel methods for snow depth measurement

**Colby College Physics Department**, Waterville, ME, USA 2017-2018
Senior Research Project, "Analyzing Model Predictions for Climate Change"

## AWARDS

**American Geophysical Union Travel Grant** 2024
Colorado State University Graduate Student Council

**Outstanding Physics Learning Assistant** 2018
American Association of Physics Teachers

**Sigma Pi Sigma** 2018
Physics Honor Society

**Distinction in Major** 2018
Physics and Mathematical Sciences, Colby College

## TEACHING EXPERIENCE

**Colorado State University**, Fort Collins, CO, USA Spring 2020
Graduate Teaching Assistant, ATS606 *Introduction to Climate*

**Colby College,** Waterville, ME, USA 2017-2018
Teaching Assistant, PH141 *Foundations of Mechanics*
Teaching Assistant, PH145 *Foundations of Electromagnetism and Optics*

## PUBLICATIONS

Díaz, L. B., A.H. Butler, D. Avisar, S. Bischof, **C.L. Boehm,** G. Boschat, W.J. Dow, C.I. Garfinkel, K.M. Grise, H. Khatri, B. Mezzina, M. Osman, J.S. Wright, P.J. Athanasiadis, J.M. Arblaster, E. Behrens, T.J. Bracegirdle, Y. Chen, E.-P. Lim, … T. Shaw, 2026: Disentangling Anthropogenic Effects on Southern Hemisphere Circulation and Surface Climate: A Multi-Model Large Ensemble Approach. *Journal of Geophysical Research: Atmospheres*, 131, e2025JD046229. https://doi.org/10.1029/2025JD046229

**Boehm, C.L.,** D.W.J. Thompson, and E. Blanchard-Wrigglesworth, 2025: The key role of the Southern Annular Mode during the sea-ice maximum for Antarctic sea ice and its recent loss. *Commun Earth Environ,* **6**, 833, https://doi.org/10.1038/s43247-025-02792-2.

**Boehm, C.L.**, and D.W.J. Thompson, 2023: The Key Role of Cloud–Climate Coupling in Extratropical Sea Surface Temperature Variability. *J. Climate*, **36**, 2753–2762, https://doi.org/10.1175/JCLI-D-22-0362.1.

## SELECTED PRESENTATIONS

**Boehm, C.L.,** Thompson, D.W.J., and E. Blanchard-Wrigglesworth, Oral Presentation Apr. 2026. "2-Way Interactions between Antarctic Sea Ice and the Atmospheric Circulation." IGS Global Seminar Series

**Boehm, C.L.,** Thompson, D.W.J., and E. Blanchard-Wrigglesworth, **Invited Oral Presentation** Sep. 2025. "The Key Role of the Southern Annular Mode during the Sea-Ice Maximum for Antarctic Sea Ice and its Recent Loss." Scientific Committee on Antarctic Research, AntClimNow (Near-term Variability and Prediction of the Antarctic Climate System) Monthly Science Talks

**Boehm, C.L.,** Thompson, D.W.J., and E. Blanchard-Wrigglesworth, Oral Presentation Jun. 2025. "Exploring the Local Atmospheric Response to Antarctic Sea Ice Loss." 30th Annual CESM Workshop 2025

**Boehm, C.L.,** Thompson, D.W.J., and E. Blanchard-Wrigglesworth, Oral Presentation Apr. 2025. "The Key Role of the Southern Annular Mode During the Seasonal Sea Ice Maximum in Recent Antarctic Sea Ice Loss." European Geosciences Union General Assembly 2025

**Boehm, C.L.,** Thompson, D.W.J., and E. Blanchard-Wrigglesworth, **Invited Oral Presentation** Mar. 2025. "Interpretation of Recent Antarctic Sea Ice Loss: The Key Role of Atmospheric Circulation During the Seasonal Sea Ice Maximum." NCAR Climate and Global Dynamics Laboratory, Climate Analysis Section Group Meeting

**Boehm, C.L.,** Thompson, D.W.J., and E. Blanchard-Wrigglesworth, Oral Presentation Mar. 2025. "Interpretation of Recent Antarctic Sea Ice Loss: The Key Role of Atmospheric Circulation During the Seasonal Sea Ice Maximum." CESM Polar Climate Working Group Meeting

**Boehm, C.L.**, Thompson, D.W.J., and E. Blanchard-Wrigglesworth, Oral Presentation Dec. 2024. "Interpretation of Recent Southern Hemisphere Sea Ice Loss: The Key Role of Atmospheric Circulation During the Seasonal Sea Ice Maximum." American Geophysical Union Annual Meeting 2024

**Boehm, C.L.**, and D.W.J. Thompson, Oral Presentation Oct. 2024. "The Key Role of Atmospheric Circulation for Recent Southern Hemisphere Sea Ice Loss." Young Scientist Symposium on Atmospheric Research

**Boehm, C.L.**, and D.W.J. Thompson, Oral Presentation Oct. 2024. "How might sea ice loss in Antarctica influence the atmospheric circulation?" Colorado State University Climate Supergroup Seminar

**Boehm, C.L.**, and D.W.J. Thompson, Oral Presentation June. 2024. "Interpretation of Recent Southern Hemisphere Sea Ice Loss: The Key Role of Atmospheric Circulation During the Seasonal Sea Ice Maximum." 24th Conference on Atmospheric and Oceanic Fluid Dynamics.

**Boehm, C.L.**, and D.W.J. Thompson, Oral Presentation Jan. 2023. "The Signature of Cloud Radiative Effects in Extratropical Sea Surface Temperature Variability." American Meteorological Society 103rd Annual Meeting

**Boehm, C.L.**, and D.W.J. Thompson, Oral Presentation Sep. 2022. "The Signature of Cloud Radiative Effects in Extratropical Sea Surface Temperature Variability." University of Bergen Stormtracks Group Meeting

**Boehm, C.L.**, and D.W.J. Thompson, Poster May. 2022. "The Contribution of Clouds to Northern Hemisphere Surface Temperature Variability on Monthly to Decadal Timescales." Stormtracks 2022: Midlatitude Storm-tracks Workshop

## SERVICE AND LEADERSHIP

**Mentor** 2023-2024
Cooperative Institute for Research in the Atmosphere/Department of Atmospheric Science Mentoring Program

**Volunteer** 2022-2024
Little Shop of Physics

**Planning Committee Co-chair** 2021-2023
Young Scientist Symposium on Atmospheric Research

**Co-founder, Member of Steering Committee** 2016-2018
Women in Physics Club at Colby College, Funded by the American Physical Society's Women in Physics Group Grant

## PROFESSIONAL DEVELOPMENT

Polar Amplification Model Intercomparison Phase 2 (PAMIP2) Workshop Oct. 2024

Polar Amplification of Climate Change Across Hemispheres and Seasons Causes and Constraints Workshop Jan. 2024

## SKILLS

Computational Languages: Python, MATLAB, LaTeX, Unix, Git, Microsoft Office